



The association between short-term exposure to ambient air pollution and fractional exhaled nitric oxide level: A systematic review and meta-analysis of panel studies[☆]

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ABSTRACT

Several epidemiological studies have evaluated the fractional exhaled nitric oxide (FeNO) of ambient air pollution but the results were controversial. We therefore conducted a systematic review and meta-analysis to investigate the associations between short-term exposure to air pollutants and FeNO level. We searched PubMed and Web of Science and included a total of 27 articles which focused on associations between ambient air pollutants (PM₁₀, PM_{2.5}, black carbon (BC), nitrogen dioxide (NO₂), sulfur dioxide (SO₂), ozone (O₃)) exposure and the change of FeNO. Random effect model was used to calculate the percent change of FeNO in association with a 10 or 1 µg/m³ increase in air pollutants exposure concentrations. A 10 µg/m³ increase in short-term PM₁₀, PM_{2.5}, NO₂, and SO₂ exposure was associated with a 3.20% (95% confidence interval (95%CI): 1.11%, 5.29%), 2.25% (95%CI: 1.51%, 2.99%), 4.90% (95%CI: 1.98%, 7.81%), and 8.28% (95%CI: 3.61%, 12.59%) change in FeNO, respectively. A 1 µg/m³ increase in short-term exposure to BC was associated with 3.42% (95%CI: 1.34%, 5.50%) change in FeNO. The association between short-term exposure to O₃ and FeNO level was insignificant (P > 0.05). Future studies are warranted to investigate the effect of multiple pollutants, different sources and composition of air pollutants on airway inflammation.

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1. Introduction

The World Health Organization (WHO) estimates that 4.2 million people worldwide die from air pollution every year and over 90% of people breathe air containing high levels of pollutants (Stanaway et al., 2018). The main sources of ambient air pollution include automobile emissions, fossil fuel combustion and industrial process, which may increase the risk of respiratory diseases including chronic obstructive pulmonary disease (COPD), asthma, lower respiratory infections and lung cancers (Brunst et al., 2015; Guan et al., 2016; Guarnieri and Balmes, 2014; Li et al., 2016).

According to the report of The Global Burden of Diseases, Injuries, and Risk Factors Study, the respiratory diseases accounted for over half of air pollution-related death in 2017, leading to 4.6 million death and 14.3 million disability-adjusted life years (DALY) globally (Stanaway et al., 2018).

FeNO is a non-invasive biomarker with excellent repeatability, high specificity and sensitivity in diagnosing, monitoring, or predicting the response to the respiratory disease (Kim et al., 2016). Several epidemiological researches applied the change of FeNO level to evaluate airway inflammation of respiratory disease caused by air pollutants. In recent years, mounting evidence suggests that short-term exposure to air pollution may be associated with the change of FeNO. However, the results were inconsistent. For example, Zhang et al. (2019) reported that a 10 µg/m³ increase in PM_{2.5} was associated with 2.59% (95%CI: 0.21%, 4.96%) increase in FeNO. Zhao et al. (2016) conducted a panel study among healthy adults indicates that a 10 µg/m³ increase in NO₂ was associated

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with 10.58% increase in FeNO. Similar results were also reported among COPD patients. Conversely, several studies have reported that air pollution was not associated with FeNO level (Adamkiewicz et al., 2004; Delfino et al., 2010; Delfino et al., 2013; Fischer et al., 2002; Liu et al., 2009; Maikawa et al., 2016; Zhao et al., 2016). A systematic review and meta-analysis can help to summarize the inconsistent evidence of those studies, however, few systematic review or meta-analysis have been published until now.

We further noticed that most of the above studies were panel study, which has a stronger design for internal validity, and was designed with repeated measurements on individual subjects at regular short-time intervals (Bloemsma et al., 2016; Modig et al., 2014; Ward and Ayres, 2004). Meanwhile, panel study can be used to explore the acute effect of air pollutants and to reveal underlying mechanism. We therefore performed a systematic review and meta-analysis to assess associations between short-term exposure to ambient air pollution and the percent change of FeNO in panel studies.

2. Methods

2.1. Search strategy

A systematic literature review from PubMed and Web of Science was conducted to identify eligible studies published in English and before September 1st 2019. The search keywords included ("Ozone" OR "black carbon" OR "diesel exhaust" OR "particulate matter" OR "outdoor exposure" OR "environmental pollution" OR "air pollution") AND ("fractional exhaled nitric oxide" OR "exhaled nitric oxide"). To avoid omission, all references were ultimately reviewed. The detailed search terms were described in supplementary materials.

2.2. Inclusion and exclusion criteria

Inclusion criteria for articles were: (1) panel studies that assessed the association between air pollution and FeNO; (2) studies conducted in all population; (3) studies estimated acute effects on FeNO level less than 14 days; (4) studies reported quantitative results such as percent change (%-change) or absolute change of FeNO. Exclusion criteria included: (1) pregnant population; (2) indoor or occupational exposure.

2.3. Quality evaluation

The quality of included studies was evaluated by Newcastle-Ottawa-Scale (NOS). The NOS contains eight items, categorized into three groups including selection, comparability and outcome. The total score ranged from 0 points to 9 points, with a higher score indicating a higher quality. In this study, we identified the study as "high quality" if NOS score ≥ 7 , otherwise, the study was regarded as "low quality". In our meta-analysis, all 27 included studies were regarded as "high quality", and the average NOS score was 7.4 for all studies (Table S2).

2.4. Data extraction

Following information from eligible studies were extracted including study design, time period, study location, participant characteristics (total number, age and health status), exposure assessment and effect estimates and 95% CI. Only quantitative results from single-pollutants model were extracted. Moreover, short-term studies usually use different lag patterns to evaluate immediate and delayed effects of air pollutants. Our meta-analysis chose the lags following the protocols summarized previously

(Yang et al., 2018). Briefly, if the study reported only one lag estimate, we extract the result directly, if multiple lags were provided, we chose one based on one of the following criteria in turn: (1) the lag that was stated priority in the study; (2) the lag was statistically significant; (3) the lag with largest effect estimate. Data extraction was performed by two independent investigators. In a case of discordance, a third reviewer was asked to give his opinion.

2.5. Quantitative data synthesis

Our meta-analysis applied percent changes (%) and 95%CI associated with a 10 $\mu\text{g}/\text{m}^3$ increases in $\text{PM}_{2.5}$, PM_{10} , SO_2 , NO_2 and O_3 . Considering the variation in daily BC concentrations is much smaller than other pollutants, we applied a 1 $\mu\text{g}/\text{m}^3$ change to estimate the effect of BC. All of the studies expressing gas pollutants (NO_2 , SO_2 , O_3) concentrations as "ppb" were converted to $\mu\text{g}/\text{m}^3$ according to $1 \text{ ppb} = M/22.4 \mu\text{g}/\text{m}^3$, where M is the molecular weight of SO_2 , NO_2 , O_3 , respectively (Liu et al., 2019a). Absolute change (not log-transformed) from linear regression models also included in our results, which were transformed back to percent changes in association with a 10 or 1 $\mu\text{g}/\text{m}^3$ increases in air pollutants: $[\beta * 10 / 1 \div M] * 100\%$, where β was absolute change and M was the mean of FeNO level (Liu et al., 2019b).

2.6. Statistical analysis

We used meta-analysis to abstract the combined estimates for the associations between air pollutants exposure and the percent change of FeNO level. Heterogeneity among studies was assessed using Cochran's Q test and quantified applying I^2 . To evaluate whether the association was modified by populations characterizes, we performed specific subgroup analysis based on several variables, including age (adult v.s. children) and geographic areas (Asia v.s. North America v.s. Europe). We also conducted sensitivity analysis of studies to check the validity of our results. Random effect model was used for all evaluated associations. Publication bias was examined using Begg's test and funnel plots. Statistical analyses were carried out using Stata software version 14.

3. Results

3.1. Study characteristics

Our initial search identified 425 studies in PubMed and Web of Science (Fig. 1). Finally, a total of 27 articles were included in our meta-analysis and they all reported the association between FeNO change and at least one of the included pollutants (Table 2). Among 27 studies, 12 were conducted in North America, 11 in Asia, and 4 in Europe (Table 1). Moreover, 8 studies reported results in the manner of absolutely change and 95%CI (Adamkiewicz et al., 2004; Carlsen et al., 2016; Delfino et al., 2006; Delfino et al., 2010; Fischer et al., 2002; Jansen et al., 2005; Jiang et al., 2019; Yoda et al., 2014) while others in percent change. Liner mixed-effect model was applied in most included studies, and only 3 studies used other models, such as the liner regression model (Karakatsani et al., 2017), generalized additive models (Liu et al., 2014) and generalized estimating equations (Lin et al., 2011). Additionally, 2 studies reported the result of asthma and non-asthma population separately, we therefore divided these 2 studies into 4 studies in our meta-analysis (Lin et al., 2011; Liu et al., 2014). A high heterogeneity existed in effect estimates between each air pollutant and FeNO (Table 2) (see Fig. 2).

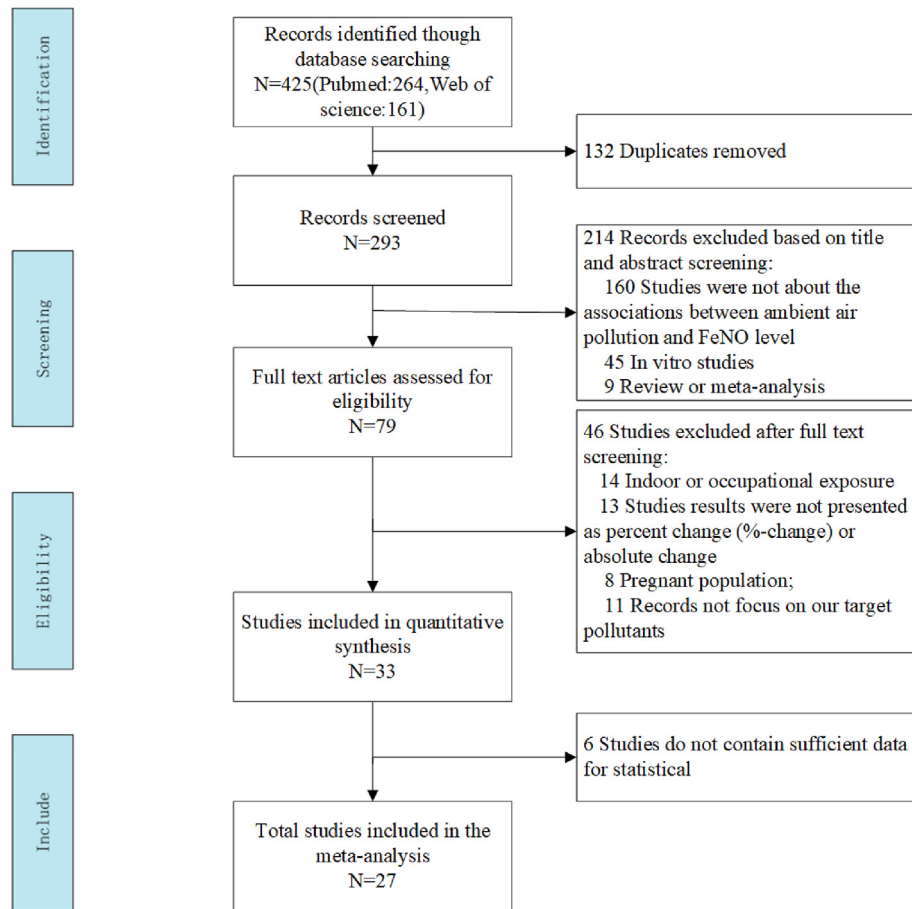


Fig. 1. Flow-chart of literature search for meta-analysis.

3.2. $PM_{2.5}$, PM_{10} and BC exposure and FeNO level

Meta-analysis showed that a $10 \mu\text{g}/\text{m}^3$ increase in PM_{10} and $PM_{2.5}$ exposure was separately associated with a 3.20% (95%CI: 1.11%, 5.29%) and 2.25% (95%CI: 1.51%, 2.99%) increase in FeNO level. As for BC exposure, the change of FeNO was 3.42% (95%CI: 1.34%, 5.50%). Subgroup analysis by geographic areas and age groups showed that the association between FeNO level and $PM_{2.5}$ was significant or stronger in Asians and adults compared with associations in North American and children (Table S3). For example, the association was significant in Asia (3.44%, 95%CI: 0.02%, 6.86%) and European (7.15%, 95%CI: 1.45%, 12.85%) compared with those in North America (3.15%, 95%CI: 3.62%, 9.92%).

3.3. Gaseous pollutants exposure and FeNO level

Positive associations were observed between NO_2 , SO_2 short-term exposure and FeNO level. Each $10 \mu\text{g}/\text{m}^3$ increase in NO_2 exposure was associated with a 4.90% (95%CI: 1.98%, 7.81%) increase in FeNO. For SO_2 , the change of FeNO was 8.28% (95%CI: 3.61%, 12.95%). However, we found no significant associations between O_3 exposure and FeNO (−0.18%, 95%CI: 1.50%, 1.14%). Subgroup analysis showed that a $10 \mu\text{g}/\text{m}^3$ increase in NO_2 was associated with a 9.77% increase in FeNO level (95%CI: 8.17%, 11.38%) in Asia, and a significant increase in FeNO among children (6.04%, 95%CI: 2.53%, 9.55%). Furthermore, there were insignificant associations between short-term exposure to O_3 and FeNO among all subgroups which were consistent with the combined results.

3.4. Sensitivity analyses and public bias

We performed sensitivity analyses to assess the stability of the results by recalculating the percent change in pooling after omitting one study each time. We found no significant change in the combined percentage change in FeNO in each group, which indicates that our results are stable and reliable. The results of Begg's test and funnel plot indicated that there was no publication bias among different air pollutants studies (Table S4).

4. Discussion

Our meta-analysis showed that a $10 \mu\text{g}/\text{m}^3$ increase in short-term exposure to $PM_{2.5}$, PM_{10} , NO_2 and SO_2 was associated with 2.25%, 3.20%, 4.90%, 8.28% increase in FeNO level, respectively. A $1 \mu\text{g}/\text{m}^3$ increase in short-term exposure to BC was associated with 3.42% (95%CI: 1.34%, 5.50%) change in FeNO. However, the association between short-term exposure to O_3 and FeNO level was insignificant.

Previous meta-analyses indicated that increased exposure to air pollution was positively associated with the development of respiratory diseases. For example, Khreis et al. (2017) reported that asthma onset was significantly associated with a $1 \mu\text{g}/\text{m}^3$ increment in $PM_{2.5}$ (OR = 1.03, 95%CI: 1.01, 1.05), a $2 \mu\text{g}/\text{m}^3$ increment in PM_{10} (OR = 1.05, 95%CI: 1.02, 1.08), and a $4 \mu\text{g}/\text{m}^3$ increment in NO_2 (OR = 1.05, 95%CI: 1.02, 1.07). Our meta-analysis indicates that short-term exposure to high concentrations of air pollutants is associated with increased FeNO levels. These results are consistent

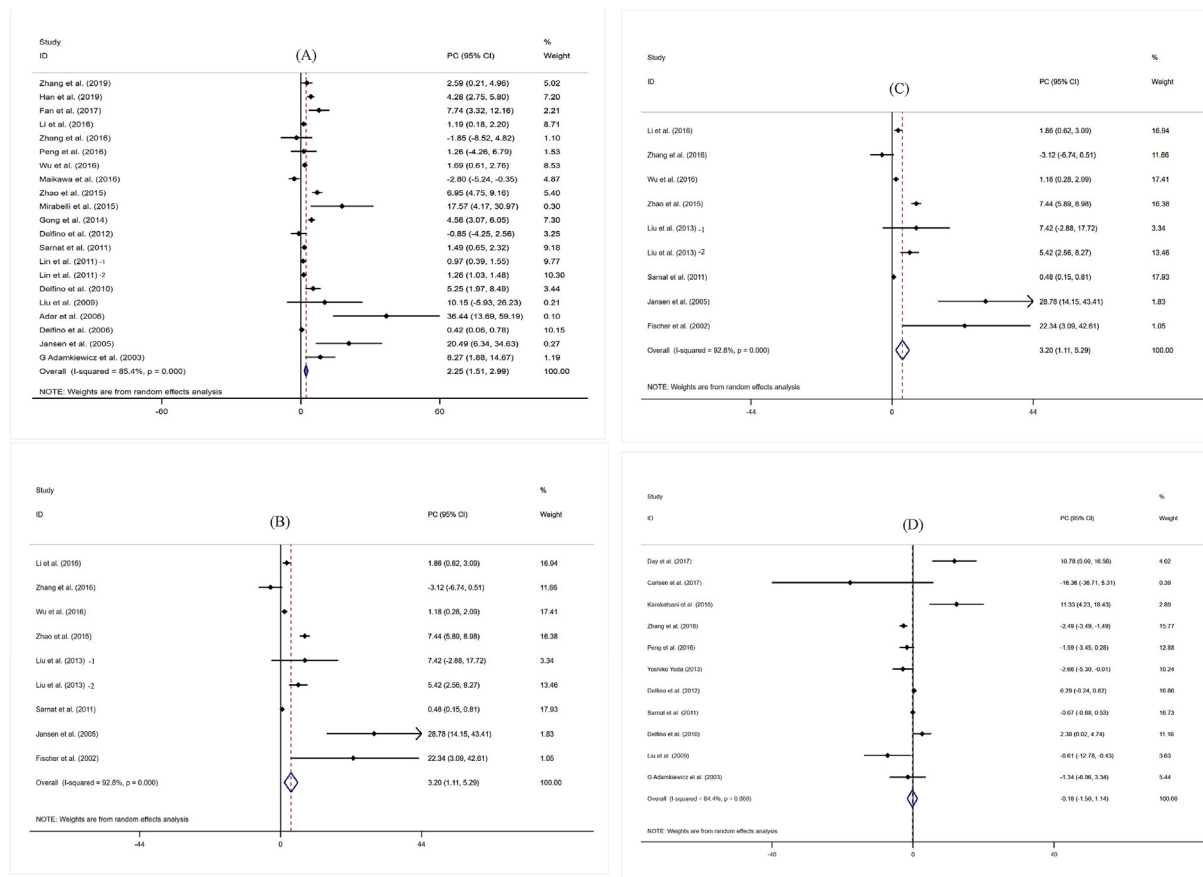


Fig. 2. Forest plot for the pooled associations between air pollutants (PM_{2.5}, PM₁₀, NO₂ and O₃) and FeNO.

with a previous review that demonstrated that increased FeNO levels were associated with an increase in a number of traffic-related air pollutants. (Scarpa et al., 2014). Therefore, we hypothesize that short-term exposure to air pollution may increase respiratory diseases through airway inflammation.

It is worth noting that PM is a mixture of heterogeneous components and PM with different size may have different physicochemical and toxicological properties. Previous research indicated PM_{2.5} is probably more harmful on respiratory system than PM₁₀ (MacIntyre et al., 2014). However, in our study, short-term exposure to PM_{2.5} (2.25%, 95%CI: 1.51%, 2.99%) had a similar effect with PM₁₀ (3.20%, 95%CI: 1.11%, 5.29%) on FeNO in each 10 µg/m³ increase. Therefore, the respiratory effect of different sizes of particles on human should be further confirmed.

We observed that short-term exposure to PM_{2.5}, PM₁₀ and BC was positively associated with FeNO. Nitric oxide (NO), which can be noninvasively measured by fractional nitric oxide concentration in exhaled breathe, is produced by a wide variety of cell types and is generated through the amino acid L-arginine by nitric oxide synthase (NOS) (Ricciardolo et al., 2004). NOS exists in three distinct isoforms: neuronal NOS (nNOS), inducible NOS (iNOS), and endothelial NOS (eNOS). Among them, NO derived from iNOS appears to be a proinflammatory mediator and has been implicated in inflammatory airway diseases. Inhaled PM may penetrate the epithelial cells lining, enhance the oxidation of L-arginine and activate the lipid peroxidation (Adar et al., 2007; Kim et al., 2016). And then the production of NO under oxidative stress conditions secondarily generates strong oxidizing agents which may modulate the development of chronic inflammatory airway disease and/or

amplify the inflammatory response (Adamkiewicz et al., 2004; Ricciardolo, 2014).

The results of this meta-analysis has shown that the percent change of FeNO was significantly associated with short-term exposure to NO₂ and SO₂ in each 10 µg/m³ increment. The underlying mechanism between FeNO and gaseous pollutants is likely modulated by DNA methylation in the arginase-nitric oxide synthase pathway (Breton et al., 2011). Cytological study on human vascular endothelial cells indicated that the hypomethylation of NOS2A could lead to increased FeNO by up-regulating iNOS, hypomethylation of ARG2 could also enhance the FeNO level by increase the L-arginase availability of iNOS (Chan et al., 2005). Meanwhile, a longitudinal panel study reported that short-term exposure to NO₂ is associated with decreased NOS2A methylation (0.19, 95%CI: 0.01, 0.37), increased ARG2 methylation (0.21, 95%CI: 0.09, 0.34) and increased FeNO (2.82%, 95%CI: 0.26%, 5.45%), respectively (Jiang et al., 2019). These studies indicated that gaseous pollutants could potentially influence FeNO level through DNA methylation, such as NOS2A and ARG2.

Subgroup analysis by geographic areas indicated that the associations between FeNO levels and a 10 µg/m³ increase in PM_{2.5} and NO₂ was generally significant or stronger in studies from Asia compared to those in North American and Europe. One probable reason for this was that the level of air pollutants in Europe and North American remained low and there was no significant fluctuation, however, in Asia, the exposure levels were higher than other regions which can vary quite dramatically (Edgington et al., 2019).

Our meta-analysis has several limitations. Firstly, the number of

Table 1

Characteristics of studies investigating the association of short-term exposure to ambient air pollution and FeNO.

Authors/year	Region	Study period	Population (n)	Age (mean $\pm$ SE or range)	Pollutants	Outcome percent changes (%) and 95%CI ^a	Covariates
Zhang et al. (2019)	China	2016	Healthy adults (43)	24.5 $\pm$ 1.5	PM _{2.5}	PM _{2.5} : 2.59 (0.21, 4.96)	Age, gender, BMI, temperature, humidity, DOW
Han et al. (2019)	China	2013–2015	General adults (120)	NA	PM _{2.5}	PM _{2.5} : 4.28 (2.74, 5.80)	Age, gender, BMI, temperature, RH, smoking status, medication usage, disease history, DOW
Jiang et al. (2019)	China	2014	Healthy adults (40)	24.5	NO ₂	NO ₂ : 2.21 (−6.09, 10.51)	Age, gender, BMI, healthy status, temperature, RH
Fan et al. (2018)	China	2014–2015	Healthy adults (46)	25 $\pm$ 3	PM _{2.5}	PM _{2.5} : 7.74 (3.32, 12.16)	Age, gender, BMI, education level, occupation, household income, RH, DOW
Karakatsani et al. (2017)	Greece	2013–2014	Healthy children (188)	10.3 $\pm$ 0.3	O ₃	O ₃ : 11.33 (4.23, 18.43)	Sex, city, area, father's education, temperature, consumption of antioxidant foods, time spent outdoors and medication usage
Day et al. (2017)	China	2014–2015	Healthy adults (89)	31.5 $\pm$ 7.6	O ₃	O ₃ : 10.78 (5.00, 16.56)	Temperature, time spent in a room with a smoker, respiratory infection status, menstruation status, hours since the participant ate and DOW
Li et al. (2017)	China	2013	Diabetic adults (35)	65.9 $\pm$ 8.4	PM _{2.5} PM ₁₀	PM _{2.5} : 1.19 (0.18, 2.20) PM ₁₀ : 1.86 (0.62, 3.09)	Age, gender, BMI, duration of diabetes, temperature, RH, DOW
Carlsen et al. (2016)	Swedish	2011	Healthy children (95)	11–12	O ₃	O ₃ : 16.36 (−36.71, 5.31)	Sex, allergy, pollen, DOW and meteorological variables
Zhang et al. (2016)	USA	2012–2014	General elderly nonsmoking adults (97)	74.8 $\pm$ 7.5	PM _{2.5} , PM ₁₀ , BC, O ₃	PM _{2.5} : 1.85 (−8.52, 4.82), PM ₁₀ : 3.12 (−6.74, 0.51) BC: 66.97 (34.33, 99.62) O ₃ : 2.49 (−3.49, −1.49)	Heat index, days of gas stove use per week
Peng et al. (2016)	USA	2006–2010	Diabetic adults (70)	64.2	PM _{2.5} O ₃	PM _{2.5} : 1.26 (−4.26, 6.79) O ₃ : 1.59 (−3.45, −1.49)	Ambient temperature, absolute humidity, season and scrubbed room NO levels
Wu et al. (2016)	China	2014	COPD adults (23)	73.6 $\pm$ 4.7	PM _{2.5} PM ₁₀ SO ₂	PM _{2.5} : 1.69 (0.61, 2.76) PM ₁₀ : 1.18 (0.28, 2.09) SO ₂ : 6.68 (3.55, 9.82)	Age, gender, BMI, temperature, humidity, DOW, a binary variable of daily timing
Maikawa et al. (2016)	Canada	2009–2010	Asthma children (62)	10.0	PM _{2.5}	PM _{2.5} : 2.80 (−5.24, −0.35)	Temperature, RH, sex, corticosteroid use, use of rescue medication, presence of allergies, occurrence of an asthma attack in the past year, eczema before the age of 2 years, and parental asthma.
Zhao et al. (2016)	China	2013	Healthy adults (75)	20–29	PM _{2.5} PM ₁₀ NO ₂ SO ₂	PM _{2.5} : 6.95 (4.75, 9.16) PM ₁₀ : 7.44 (5.89, 8.98) NO ₂ : 10.58 (6.97, 14.18) SO ₂ : 58.33 (43.58, 73.08)	The moving average of temperature and RH on the current day
Mirabelli et al. (2015)	USA	2009–2011	Asthma adults	31.0	PM _{2.5}	PM _{2.5} : 17.57 (4.17, 30.97)	Age, race, gender
De Prins et al. (2014)	Western Europe	2011	Healthy children (130)	8.9 $\pm$ 1.4	BC	BC: 47.02 (4.95, 89.09)	Age, gender, parents' education level, airway allergy, temperature, RH
Yoda et al. (2014)	Japan	2012	Healthy adults (21)	21 $\pm$ 0.6	NO ₂	O ₃ : 2.66 (−5.30, −0.01)	Temperature, RH
Gong et al. (2014)	China	2008	Healthy adults (125)	22–27	PM _{2.5}	PM _{2.5} : 4.56 (3.07, 6.05)	Temperature, RH, gender, DOW
Delfino et al. (2013)	USA	2003–2004	Asthma children (45)	13.5	PM _{2.5} NO ₂ O ₃	PM _{2.5} : 0.85 (−4.25, 2.56) NO ₂ : 0.02 (−0.95, 0.90) O ₃ : 0.29 (−0.24, 0.82)	Temperature, RH
Sarnat et al. (2012)	U.S. –Mexico border	2008	Asthma children (58)	8.7	PM _{2.5} PM ₁₀ BC NO ₂ O ₃	PM _{2.5} : 1.49 (0.65, 2.32) PM ₁₀ : 0.48 (0.15, 0.81) BC: 4.17 (−7.72, 16.05)	School, ambient temperature, RH

(continued on next page)

Table 1 (continued)

Authors/year	Region	Study period	Population (n)	Age (mean ± SE or range)	Pollutants	Outcome percent changes (%) and 95%CI ^a	Covariates
Lin et al. (2011)	China	2008	General children (36) Asthma and health	10.6	PM _{2.5} BC NO ₂ SO ₂	NO ₂ : 1.48 (0.59, 2.38) O ₃ : 0.07 (−0.68, 0.53) PM _{2.5} : 1.16 (0.94, 1.37) BC: 46.81 (38.84, 54.79) NO ₂ : 9.84 (8.35, 11.34) SO ₂ : 6.59 (5.30, 7.89)	Temperature, RH, BMI, and asthma (yes or no).
Delfino et al. (2010)	USA	2005–2007	Nonsmoking elderly adults with confirmed history of coronary (60)	84.1 ± 5.6	PM _{2.5}	PM _{2.5} : 5.25 (1.97, 8.49)	Seasonal phase and community
Liu et al. (2009)	Canada	2005	Asthma children (182)	11.0	PM _{2.5} NO ₂ O ₃ SO ₂	PM _{2.5} : 10.15 (−5.93, 26.23) NO ₂ : 4.16 (−1.45, 9.77) O ₃ : 6.61 (−12.78, −0.43) SO ₂ : 3.25 (−2.72, 9.21)	Testing period, sex and daily use of inhaled corticosteroids
Adar et al. (2007)	USA	2002	General adults (44)	62–94	PM _{2.5}	PM _{2.5} : 36.44 (13.69, 59.19)	DOW, trip collection device, vitamin use, pollen, mold, temperature during the current hour, time to analysis, and nitric oxide concentrations in the testing room
Delfino et al. (2006)	USA	2003	Asthma children (45)	13.5	PM _{2.5}	PM _{2.5} : 0.42 (0.06, 0.78)	Respiratory infections, region of study, sex, cumulative daily use of as-needed β-agonist inhalers, and weekend
Jansen et al. (2005)	USA	2002–2003	Elderly adults with asthma or COPD	75	PM _{2.5} PM ₁₀ BC	PM _{2.5} : 20.49 (6.34, 34.63) PM ₁₀ : 28.78 (14.15, 43.41) BC: 58.54 (9.76, 107.32)	Age, RH, and temperature.
Adamkiewicz et al. (2004)	USA	2000	Non-smoking adults (29)	70.7	PM _{2.5} NO ₂ O ₃ SO ₂	PM _{2.5} : 8.27 (1.88, 14.67) NO ₂ : 3.58 (−2.36, 9.52) O ₃ : 1.34 (−6.06, 3.34) SO ₂ : 0.74 (−1.87, 3.38)	Time of day, day of the week, and the week of study
Fischer et al. (2002)	Netherlands	NA	General children (68)	10.4	PM ₁₀ NO ₂	PM ₁₀ : 22.34 (3.09, 42.61) NO ₂ : 20.96 (−19.59, 66.32)	Temperature

NA: Not reported; BMI: body mass index; RH: relative humidity; DOW: day of week.
^a Percent change of FeNO in association with 10 µg/m³ increase in ambient air pollution.

included studies was relatively few, especially for BC and SO₂, which stop us to conduct subgroup analysis. Meanwhile, the non-significant Begg’s tests are very likely due to the small number of studies and consequently low statistical power. Secondly, though subgroup analysis by geographic areas and age groups were

conducted, we did not find the sources of heterogeneity successfully. The healthy status of the included population, which contain general population and different types of patients, might be an important source of heterogeneity among studies. However, the few number (n < 5) of included studies prevented us from

Table 2
 Meta-analysis of percent change in association with a 10/1 µg/m³ increase in ambient air pollution concentrations.

Pollutants	Overall analysis		No. of effect estimates	No. of studies	Heterogeneity	
	Pooled %-changes (95%CI)	Significance test			P-value	I ²
PM _{2.5}	2.25 (1.51, 2.99)	P < 0.001	21	20	P < 0.001	85.4%
PM ₁₀	3.20 (1.11, 5.29)	P = 0.003	9	8	P < 0.001	92.8%
BC	3.42 (1.34, 5.50)	P < 0.001	6	5	P < 0.001	88.0%
NO ₂	4.90 (1.98, 7.81)	P = 0.001	11	10	P < 0.001	93.3%
O ₃	−0.18 (−1.50, 1.14)	P = 0.787	11	11	P < 0.001	84.4%
SO ₂	8.28 (3.61, 12.59)	P = 0.010	6	5	P < 0.001	93.9%

conducting subgroup analyses. Thirdly, we did not conduct the subgroup analysis by lag structure because different studies showed results with different lags. Finally, results based on multiple-pollutant model were not considered in our meta-analysis due to it is difficult to implement and validate. Therefore, we only estimated the associations between each pollutant and FeNO level based on single-pollutant model results and did not evaluate the interactions between the air pollutants.

5. Conclusion

High concentration of short-term exposure to air pollution was associated with increased FeNO, suggesting that air pollution exposure may increase respiratory diseases through airway inflammation. Future studies should consider interactions between pollutants and explore the source-specific effects of air pollutants. Meanwhile, more studies should be conducted to explore the specific mechanism behind the health effect of short-term exposure to each air pollutant.

Declaration of competing interest

All authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2020.114833>.

References

- Adamkiewicz, G., Ebelt, S., Syring, M., Slater, J., Speizer, F.E., Schwartz, J., et al., 2004. Association between air pollution exposure and exhaled nitric oxide in an elderly population. *Thorax* 59, 204–209.
- Adar, S.D., Adamkiewicz, G., Gold, D.R., Schwartz, J., Coull, B.A., Suh, H., 2007. Ambient and microenvironmental particles and exhaled nitric oxide before and after a group bus trip. *Environ. Health Perspect.* 115, 507–512.
- Bloemsma, L.D., Hoek, G., Smit, L.A.M., 2016. Panel studies of air pollution in patients with copd: systematic review and meta-analysis. *Environ. Res.* 151, 458–468.
- Breton, C.V., Byun, H.M., Wang, X., Salam, M.T., Siegmund, K., Gilliland, F.D., 2011. DNA methylation in the arginase-nitric oxide synthase pathway is associated with exhaled nitric oxide in children with asthma. *Am. J. Respir. Crit. Care Med.* 184, 191–197.
- Brunst, K.J., Ryan, P.H., Brokamp, C., Bernstein, D., Reponen, T., Lockey, J., et al., 2015. Timing and duration of traffic-related air pollution exposure and the risk for childhood wheeze and asthma. *Am. J. Respir. Crit. Care Med.* 192, 421–427.
- Carlsen, H.K., Boman, P., Bjor, B., Olin, A.C., Forsberg, B., 2016. Coarse fraction particle matter and exhaled nitric oxide in non-asthmatic children. *Int. J. Environ. Res. Publ. Health* 13.
- Chan, G.C., Fish, J.E., Mawji, I.A., Leung, D.D., Rachlis, A.C., Marsden, P.A., 2005. Epigenetic basis for the transcriptional hyporesponsiveness of the human inducible nitric oxide synthase gene in vascular endothelial cells. *J. Immunol.* 175, 3846–3861.
- Day, D.B., Xiang, J., Mo, J., Li, F., Chung, M., Gong, J., et al., 2017. Association of ozone exposure with cardiorespiratory pathophysiological mechanisms in healthy adults. *JAMA Int. Med.* 177, 1344–1353.
- De Prins, S., Dons, E., Van Poppel, M., Int Panis, L., Van de Mieroop, E., Nelen, V., et al., 2014. Airway oxidative stress and inflammation markers in exhaled breath from children are linked with exposure to black carbon. *Environ. Int.* 73, 440–446.
- Delfino, R.J., Staimer, N., Gillen, D., Tjoa, T., Sioutas, C., Fung, K., et al., 2006. Personal and ambient air pollution is associated with increased exhaled nitric oxide in children with asthma. *Environ. Health Perspect.* 114, 1736–1743.
- Delfino, R.J., Staimer, N., Tjoa, T., Arhami, M., Polidori, A., Gillen, D.L., et al., 2010. Associations of primary and secondary organic aerosols with airway and systemic inflammation in an elderly panel cohort. *Epidemiology* 21, 892–902.
- Delfino, R.J., Staimer, N., Tjoa, T., Gillen, D.L., Schauer, J.J., Shafer, M.M., 2013. Airway inflammation and oxidative potential of air pollutant particles in a pediatric asthma panel. *J. Expo. Sci. Environ. Epidemiol.* 23, 466–473.
- Edgington, S., O'Sullivan, D.E., King, W., Loughheed, M.D., 2019. Effect of outdoor particulate air pollution on fev1 in healthy adults: a systematic review and meta-analysis. *Occup. Environ. Med.* 76, 583–591.
- Fan, Z., Pun, V.C., Chen, X.C., Hong, Q., Tian, L., Ho, S.S., et al., 2018. Personal exposure to fine particles (pm2.5) and respiratory inflammation of common residents in Hong Kong. *Environ. Res.* 164, 24–31.
- Fischer, P.H., Steerenberg, P.A., Snelder, J.D., van Loveren, H., van Amsterdam, J.G., 2002. Association between exhaled nitric oxide, ambient air pollution and respiratory health in school children. *Int. Arch. Occup. Environ. Health* 75, 348–353.
- Gong, J., Zhu, T., Kipen, H., Wang, G., Hu, M., Guo, Q., et al., 2014. Comparisons of ultrafine and fine particles in their associations with biomarkers reflecting physiological pathways. *Environ. Sci. Technol.* 48, 5264–5273.
- Guan, W.-J., Zheng, X.-Y., Chung, K.F., Zhong, N.-S., 2016. Impact of air pollution on the burden of chronic respiratory diseases in China: time for urgent action. *Lancet* 388, 1939–1951.
- Guarnieri, M., Balmes, J.R., 2014. Outdoor air pollution and asthma. *Lancet* 383, 1581–1592.
- Han, Y., Wang, Y., Li, W., Chen, X., Xue, T., Chen, W., et al., 2019. Susceptibility of prediabetes to the health effect of air pollution: a community-based panel study with a nested case-control design. *Environ. Health* 18, 65.
- Jansen, K.L., Larson, T.V., Koenig, J.Q., Mar, T.F., Fields, C., Stewart, J., et al., 2005. Associations between health effects and particulate matter and black carbon in subjects with respiratory disease. *Environ. Health Perspect.* 113, 1741–1746.
- Jiang, Y., Niu, Y., Xia, Y., Liu, C., Lin, Z., Wang, W., et al., 2019. Effects of personal nitrogen dioxide exposure on airway inflammation and lung function. *Environ. Res.* 177, 108620.
- Karakatsani, A., Samoli, E., Rodopoulou, S., Dimakopoulou, K., Papakosta, D., Spyrtos, D., et al., 2017. Weekly personal ozone exposure and respiratory health in a panel of Greek schoolchildren. *Environ. Health Perspect.* 125, 077016.
- Khreis, H., Kelly, C., Tate, J., Parslow, R., Lucas, K., Nieuwenhuijsen, M., 2017. Exposure to traffic-related air pollution and risk of development of childhood asthma: a systematic review and meta-analysis. *Environ. Int.* 100.
- Kim, H.B., Eckel, S.P., Kim, J.H., Gilliland, F.D., 2016. Exhaled no: determinants and clinical application in children with allergic airway disease. *Allergy Asthma Immunol. Res.* 8, 12–21.
- Li, H., Bai, H., Yang, C., Chen, R., Wang, C., Zhao, Z., et al., 2017. Acute effects of ambient temperature and particulate air pollution on fractional exhaled nitric oxide: a panel study among diabetic patients in shanghai, China. *J. Epidemiol.* 27, 584–589.
- Li, M.-H., Fan, L.-C., Mao, B., Yang, J.-W., Choi, A.M.K., Cao, W.-J., et al., 2016. Short-term exposure to ambient fine particulate matter increases hospitalizations and mortality in copd: a systematic review and meta-analysis. *Chest* 149, 447–458.
- Lin, W., Huang, W., Zhu, T., Hu, M., Brunekreef, B., Zhang, Y., et al., 2011. Acute respiratory inflammation in children and black carbon in ambient air before and during the 2008 beijing olympics. *Environ. Health Perspect.* 119, 1507–1512.
- Liu, C., Flexeder, C., Fuertes, E., Cyrys, J., Bauer, C.P., Koletzko, S., et al., 2014. Effects of air pollution on exhaled nitric oxide in children: results from the giniplus and lisaplus studies. *Int. J. Hyg Environ. Health* 217, 483–491.
- Liu, F., Chen, G., Huo, W., Wang, C., Liu, S., Li, N., et al., 2019a. Associations between long-term exposure to ambient air pollution and risk of type 2 diabetes mellitus: a systematic review and meta-analysis. *Environ. Pollut.* 252, 1235–1245.
- Liu, L., Poon, R., Chen, L., Frescura, A.M., Montuschi, P., Ciabattini, G., et al., 2009. Acute effects of air pollution on pulmonary function, airway inflammation, and oxidative stress in asthmatic children. *Environ. Health Perspect.* 117, 668–674.
- Liu, Q., Gu, X., Deng, F., Mu, L., Baccarelli, A.A., Guo, X., et al., 2019b. Ambient particulate air pollution and circulating c-reactive protein level: a systematic review and meta-analysis. *Int. J. Hyg Environ. Health* 222, 756–764.
- MacIntyre, E.A., Gehring, U., Mölter, A., Fuertes, E., Klümper, C., Krämer, U., et al., 2014. Air pollution and respiratory infections during early childhood: an analysis of 10 european birth cohorts within the escape project. *Environ. Health Perspect.* 122, 107–113.
- Maikawa, C.L., Weichenthal, S., Wheeler, A.J., Dobbin, N.A., Smargiassi, A., Evans, G., et al., 2016. Particulate oxidative burden as a predictor of exhaled nitric oxide in children with asthma. *Environ. Health Perspect.* 124, 1616–1622.
- Mirabelli, M.C., Golan, R., Greenwald, R., Raysoni, A.U., Holguin, F., Kewada, P., et al., 2015. Modification of traffic-related respiratory response by asthma control in a population of car commuters. *Epidemiology* 26, 546–555.
- Modig, L., Dahgam, S., Olsson, D., Nyberg, F., Wass, K., Forsberg, B., et al., 2014. Short-term exposure to ozone and levels of exhaled nitric oxide. *Epidemiology* 25, 79–87.

- Peng, C., Luttmann-Gibson, H., Zanobetti, A., Cohen, A., De Souza, C., Coull, B.A., et al., 2016. Air pollution influences on exhaled nitric oxide among people with type ii diabetes. *Air Qual. Atmos. Health* 9, 265–273.
- Ricciardolo, F.L., 2014. Revisiting the role of exhaled nitric oxide in asthma. *Curr. Opin. Pulm. Med.* 20, 53–59.
- Ricciardolo, F.L.M., Sterk, P.J., Gaston, B., Folkerts, G., 2004. Nitric oxide in health and disease of the respiratory system. *Physiol. Rev.* 84, 731–765.
- Sarnat, S.E., Raysoni, A.U., Li, W.W., Holguin, F., Johnson, B.A., Flores Luevano, S., et al., 2012. Air pollution and acute respiratory response in a panel of asthmatic children along the u.S.-Mexico border. *Environ. Health Perspect.* 120, 437–444.
- Scarpa, M.C., Kulkarni, N., Maestrelli, P., 2014. The role of non-invasive biomarkers in detecting acute respiratory effects of traffic-related air pollution. *Clin. Exp. Allergy* 44, 1100–1118.
- Stanaway, J.D., Afshin, A., Gakidou, E., Lim, S.S., Abate, D., Abate, K.H., et al., 2018. Global, regional, and national comparative risk assessment of 84 behavioural, environmental and occupational, and metabolic risks or clusters of risks for 195 countries and territories, 1990–2017: a systematic analysis for the global burden of disease study 2017. *Lancet* 392, 1923–1994.
- Ward, D.J., Ayres, J.G., 2004. Particulate air pollution and panel studies in children: a systematic review. *Occup. Environ. Med.* 61, e13.
- Wu, S., Ni, Y., Li, H., Pan, L., Yang, D., Baccarelli, A.A., et al., 2016. Short-term exposure to high ambient air pollution increases airway inflammation and respiratory symptoms in chronic obstructive pulmonary disease patients in Beijing, China. *Environ. Int.* 94, 76–82.
- Yang, B.Y., Qian, Z., Howard, S.W., Vaughn, M.G., Fan, S.J., Liu, K.K., et al., 2018. Global association between ambient air pollution and blood pressure: a systematic review and meta-analysis. *Environ. Pollut.* 235, 576–588.
- Yoda, Y., Otani, N., Sakurai, S., Shima, M., 2014. Acute effects of summer air pollution on pulmonary function and airway inflammation in healthy young women. *J. Epidemiol.* 24, 312–320.
- Zhang, Q., Wang, W., Niu, Y., Xia, Y., Lei, X., Huo, J., et al., 2019. The effects of fine particulate matter constituents on exhaled nitric oxide and DNA methylation in the arginase-nitric oxide synthase pathway. *Environ. Int.* 131, 105019.
- Zhang, X., Staimer, N., Gillen, D.L., Tjoa, T., Schauer, J.J., Shafer, M.M., et al., 2016. Associations of oxidative stress and inflammatory biomarkers with chemically-characterized air pollutant exposures in an elderly cohort. *Environ. Res.* 150, 306–319.
- Zhao, Z., Chen, R., Lin, Z., Cai, J., Yang, Y., Yang, D., et al., 2016. Ambient carbon monoxide associated with alleviated respiratory inflammation in healthy young adults. *Environ. Pollut.* 208, 294–298.